

Technical Document #946: Natural Language Processing - Comprehensive Overview

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Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.
- Machine translation converts text from one language to another.
- Text summarization generates a concise summary of a longer text.